

Opposing moment



- Null-space projector

$$N_\tau = I_7 - J^\top J^{+\top}, \quad \tau_{\text{null}} = \tau_d + \tau_\sigma$$

Projects secondary joint torques into the kinematic null space.

J : joint $\rightarrow$ EE velocities

J^+ : pseudoinverse

I_7 : 7×7 identity matrix

- Damping torque

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

Opposes redundant joint velocity.
Zero at zero joint velocity.

d_{null} : damping coefficient [N m s/rad]

$\dot{q}$: measured joint velocity

- Conditioning torque

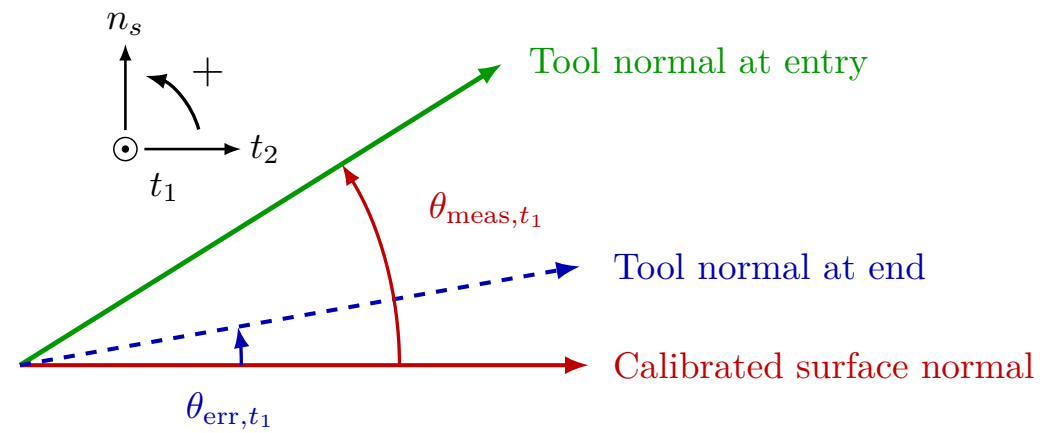
$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

Seeks better local conditioning.
Can act at zero joint velocity.

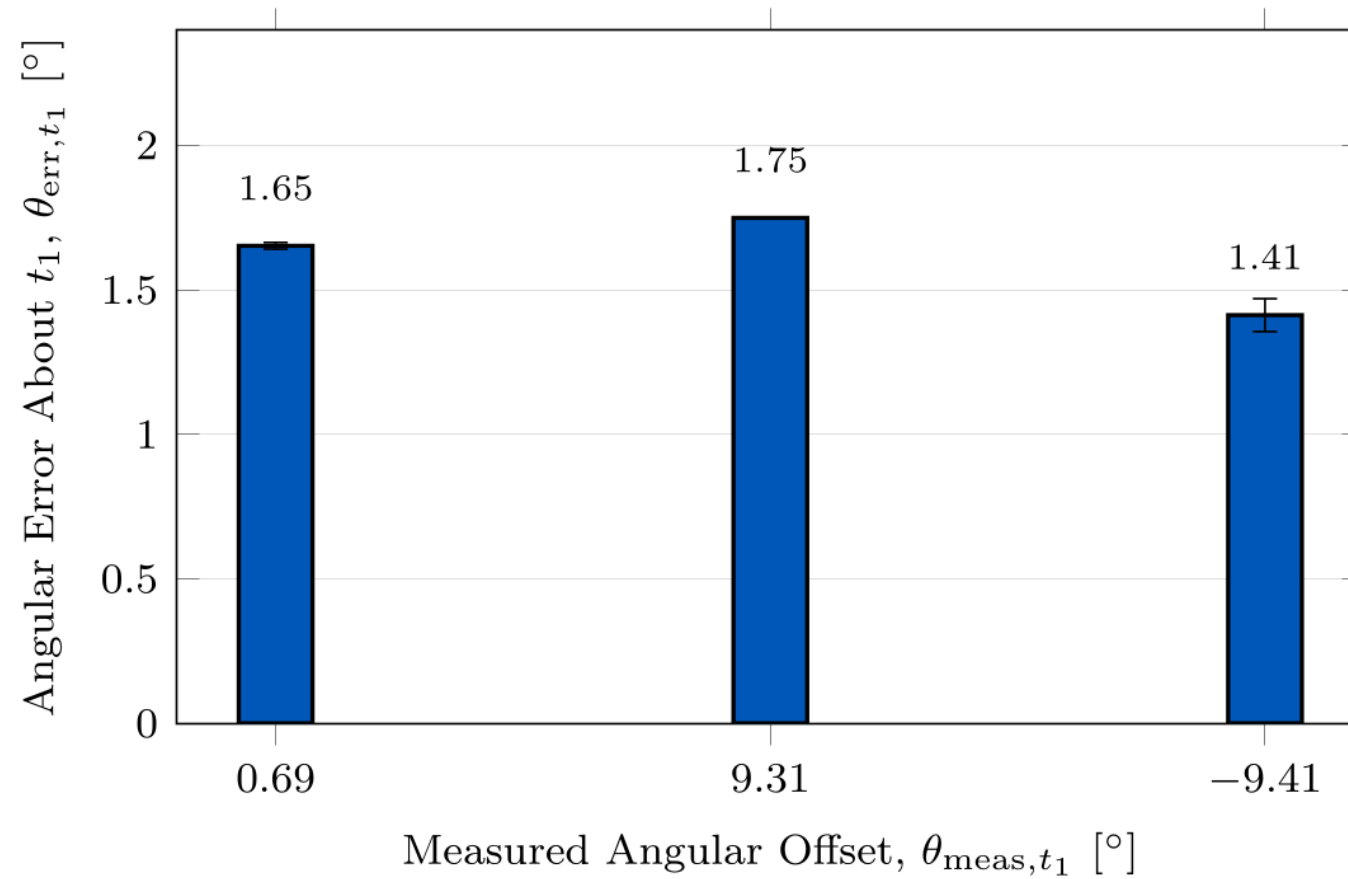
k_σ : torque magnitude [N m]

v_7 : unit null-space direction

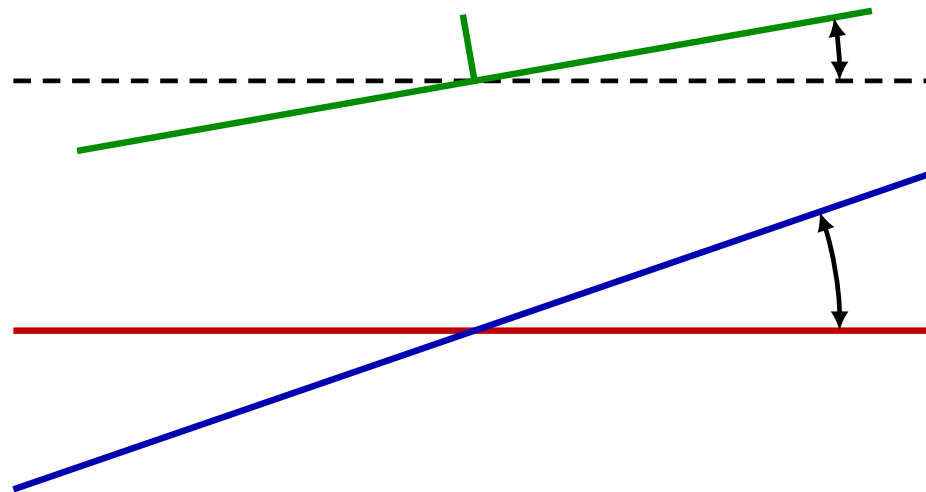
s_σ : direction choice, zero in deadband



Baseline angular error



Sources of angular offset



Desired–achieved difference

Configured–physical difference

— Configured surface

- - - Desired orientation

— Physical surface

— Tool face